RANS PART 1: REYNOLDS DECOMPOSITION & THE MASS CONTINUITY EQUATION

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Fluid Dynamics and IBM Theory • Module 5

Adapted from a lecture by Steven L. Brunton.

Claude (Anthropic) was used to assist with formatting, organization, and clarification of the instructor's original notes.

ROADMAP (PART 1)

- Start with incompressible Navier–Stokes (nondimensional form).
- Define the Reynolds decomposition: $\boldsymbol{u}(\boldsymbol{x}, t) = \overline{\boldsymbol{u}}(\boldsymbol{x}) + \boldsymbol{u}'(\boldsymbol{x}, t)$ for a stationary mean.
- Define the time-average operator and its key properties.
- Plug decomposition into **mass continuity** $\nabla \cdot \boldsymbol{u} = 0$.
- Time-average to derive:

$$\nabla \cdot \overline{\boldsymbol{u}} = 0, \qquad \nabla \cdot \boldsymbol{u}' = 0.$$

INCOMPRESSIBLE NAVIER–STOKES (NONDIMENSIONAL)

We work with the incompressible Navier–Stokes equations:

$$\frac{\partial \boldsymbol{u}}{\partial t} + (\boldsymbol{u} \cdot \nabla)\boldsymbol{u} = -\nabla p + \frac{1}{\mathrm{Re}}\,\Delta \boldsymbol{u},$$
$$\nabla \cdot \boldsymbol{u} = 0.$$

- $\boldsymbol{u}(\boldsymbol{x}, t) \in \mathbb{R}^3$ is the velocity field.
- $p(\boldsymbol{x}, t)$ is pressure (scaled).
- Re is the Reynolds number (the ratio of inertial to viscous effects for the chosen scales).

Interpretation: first line is **momentum conservation**, second line is **mass conservation** (incompressibility).

REMINDER: $\boldsymbol{u}$ IS A 3-COMPONENT FIELD

Write

$$\boldsymbol{u}(\boldsymbol{x},t) = \begin{bmatrix} u(\boldsymbol{x},t) \\ v(\boldsymbol{x},t) \\ w(\boldsymbol{x},t) \end{bmatrix}, \qquad \boldsymbol{x} = (x,y,z).$$

Then incompressibility is

$$\nabla \cdot \boldsymbol{u} = \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0.$$

Today: we will apply Reynolds averaging to this equation.

REYNOLDS DECOMPOSITION

We decompose the flow into a mean and a fluctuation:

$$\boldsymbol{u}(\boldsymbol{x}, t) = \overline{\boldsymbol{u}}(\boldsymbol{x}) + \boldsymbol{u}'(\boldsymbol{x}, t).$$

Component-wise:

$$u = \overline{u} + u', \quad v = \overline{v} + v', \quad w = \overline{w} + w'.$$

- Mean flow $\overline{\boldsymbol{u}}(\boldsymbol{x})$ is **time-averaged**; it depends only on space under the stationarity assumption used here.
- Fluctuations $\boldsymbol{u}'(\boldsymbol{x}, t)$ capture all remaining unsteady turbulent content.

TIME AVERAGE (STATIONARY ASSUMPTION)

For a statistically stationary flow, define the long-time average at each spatial point:

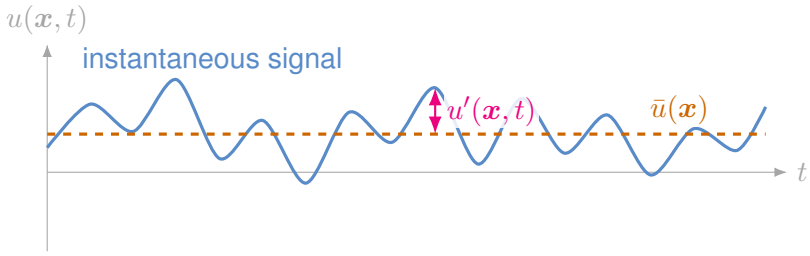
$$\overline{\boldsymbol{u}}(\boldsymbol{x}) \;:=\; \lim_{T\to\infty} \frac{1}{T} \int_0^T \boldsymbol{u}(\boldsymbol{x}, t)\, dt.$$

- We assume the relevant statistics are **stationary** and the long-time average is well defined.
- Then the fluctuation is defined by

$$\boldsymbol{u}'(\boldsymbol{x}, t) := \boldsymbol{u}(\boldsymbol{x}, t) - \overline{\boldsymbol{u}}(\boldsymbol{x}).$$

Historical motivation: point measurements (e.g., hot-wire anemometry) produce long time series at fixed $\boldsymbol{x}$. For nonstationary flows, an ensemble or finite-window average may retain time dependence.

A SIGNAL, ITS MEAN, AND ITS FLUCTUATION

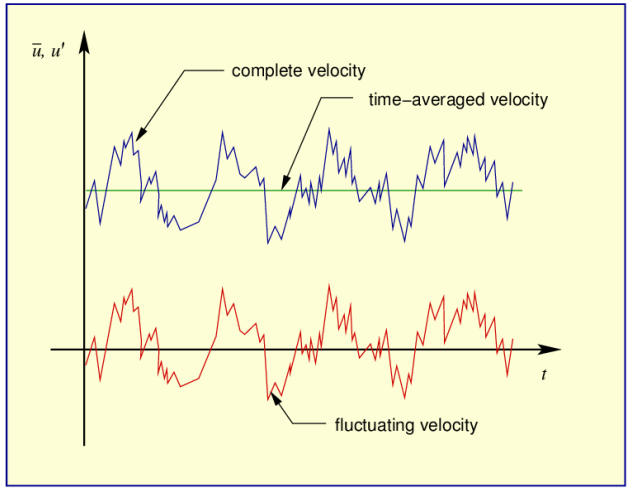


Schematic: instructor's course notes.

Pointwise identity

At every sampled time, $u(\boldsymbol{x}, t) = \bar{u}(\boldsymbol{x}) + u'(\boldsymbol{x}, t)$, while $\overline{u'} = 0$ by construction.

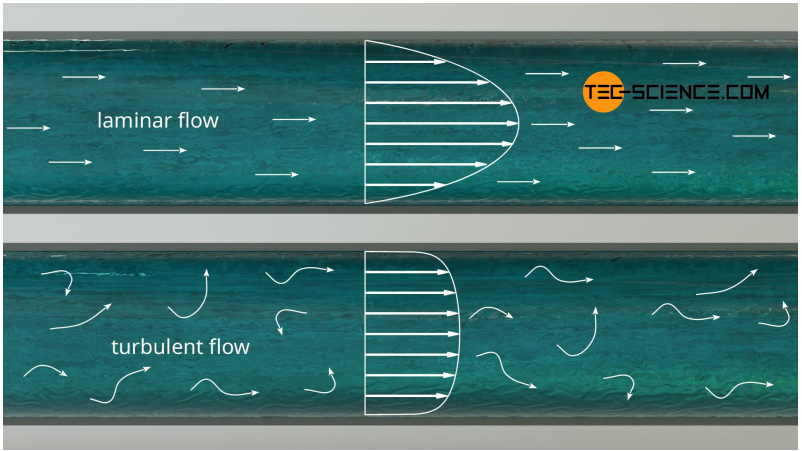
REYNOLDS DECOMPOSITION: ILLUSTRATION



Source: McDonough, James (2009). Lectures in elementary fluid dynamics: physics, mathematics and applications.

J. M. McDonough, lecture notes, University of Kentucky (2009); figure number to be confirmed.

TURBULENT PIPE FLOW



Source: https://www.tec-science.com/mechanics/gases-and-liquids/pressure-loss-in-pipe-systems/

tec-science.com, “Pressure loss in pipe systems” (educational website). Top: laminar parabolic profile. Bottom: the *time-averaged* turbulent profile is much flatter, with steep gradients near the wall; the instantaneous velocity fluctuates about it.

WHY AVERAGE? (MODEL REDUCTION VIEWPOINT)

Direct simulation of fully turbulent Navier–Stokes may require resolving:

- largest energetic scales **and** smallest dissipative scales,
- often $\Rightarrow$ millions to billions of degrees of freedom.

In many applications, we care about **mean quantities** (e.g., drag, lift, separation) and perhaps a few low-order statistics of fluctuations.

Key idea

Try to model $\overline{\boldsymbol{u}}(\boldsymbol{x})$ and selected statistics of $\boldsymbol{u}'$ without resolving every eddy.

ALGEBRA YOU NEED: PROPERTIES OF TIME AVERAGES

Let $\overline{\cdot}$ denote time averaging.

- **Zero-mean fluctuation:** $\overline{\boldsymbol{u}'} = 0$ (by definition).
- **Linearity:** $\overline{u+v} = \overline{u} + \overline{v}$.
- **Idempotent:** $\overline{\overline{u}} = \overline{u}$.
- **Commutes with differentiation:** $\overline{\dfrac{\partial a}{\partial t}} = \dfrac{\partial \overline{a}}{\partial t}$ (and likewise for spatial derivatives). For the long-time average of a bounded, stationary signal both sides vanish: $\overline{\partial_t a} = \lim_{T\to\infty} \big(a(T) - a(0)\big)/T = 0$ and $\overline{a}$ does not depend on t. For ensemble or finite-window averages the rule is nontrivial and needs enough regularity to interchange the derivative with the integral or limit.
- **Mean of mean times quantity:** $\overline{\overline{u}\cdot v} = \overline{u}\cdot\overline{v}$.
- **Mean times mean:** if $u = \overline{u}$ and $v = \overline{v}$ then $\overline{u\cdot v} = u\cdot v$.

Important warning (closure starts here)

In general,

$$\overline{u\cdot v} \neq \overline{u}\cdot\overline{v}.$$

COMMUTING THE AVERAGE WITH SPATIAL DERIVATIVES

Because we average in **time** (not space), we typically use:

$$\overline{\frac{\partial u}{\partial s}} = \frac{\partial}{\partial s}\overline{u},$$

provided we can interchange differentiation and integration (regularity assumptions).

- If we time average a derivative, it's the same as the derivative of the time average.
- This is what makes the continuity derivation clean.
- Note we are **not** averaging in space here.

VARIANCE AND HIGHER MOMENTS OF THE FLUCTUATION

Important warning

$$\overline{u'^2} \neq 0$$

- The time average of the fluctuation is zero, $\overline{\boldsymbol{u}'} = 0$.
- Its variance and higher-order moments generally are not zero.
- These higher moments are important in the dynamics of the fluid, and we will need to keep track of them.
- Index notation: write $(u_1, u_2, u_3) = (u, v, w)$ and $(x_1, x_2, x_3) = (x, y, z)$, so u_i' is the i-th fluctuation component and, e.g., $R_{12} = \overline{u'v'}$.
- $R_{ij} = \overline{u_i' u_j'}$ has units of velocity squared. The dimensional turbulent-stress contribution is conventionally $-\rho R_{ij}$.

THE HEART OF THE CLOSURE PROBLEM

- We are trying to take this very complicated equation that varies in space and time and obtain the mean flow and selected fluctuation statistics.
- But we don’t want to solve NS in space and time everywhere.
- This challenge is at the heart of the closure problem. We end up with more unknowns than equations.
- We will need to make approximations and assumptions based on physical intuition to build closure models to determine the unknowns.
- Data-driven and machine-learning models can assist with these approximations, but they must still respect the governing physics and be tested a posteriori.

STEP 1: PLUG DECOMPOSITION INTO CONTINUITY

Start with incompressibility:

$$\nabla \cdot \boldsymbol{u} = 0.$$

Substitute $\boldsymbol{u} = \overline{\boldsymbol{u}} + \boldsymbol{u}'$:

$$\nabla \cdot (\overline{\boldsymbol{u}} + \boldsymbol{u}') = 0 \quad \Rightarrow \quad \nabla \cdot \overline{\boldsymbol{u}} + \nabla \cdot \boldsymbol{u}' = 0.$$

In components:

$$\frac{\partial \overline{u}}{\partial x} + \frac{\partial \overline{v}}{\partial y} + \frac{\partial \overline{w}}{\partial z} + \frac{\partial u'}{\partial x} + \frac{\partial v'}{\partial y} + \frac{\partial w'}{\partial z} = 0.$$

STEP 2: TIME-AVERAGE THE CONTINUITY EQUATION

We can learn a lot by time averaging this equation, so take the time average of

$$\nabla \cdot \overline{\boldsymbol{u}} + \nabla \cdot \boldsymbol{u}' = 0.$$

Using linearity, commutation with spatial derivatives, idempotence ($\overline{\overline{\boldsymbol{u}}} = \overline{\boldsymbol{u}}$, so $\overline{\nabla \cdot \overline{\boldsymbol{u}}} = \nabla \cdot \overline{\boldsymbol{u}}$), and $\overline{\boldsymbol{u}'} = 0$:

$$\overline{\nabla \cdot \overline{\boldsymbol{u}}} + \overline{\nabla \cdot \boldsymbol{u}'} = 0 \quad \Rightarrow \quad \nabla \cdot \overline{\boldsymbol{u}} + \nabla \cdot \overline{\boldsymbol{u}'} = 0 \quad \Rightarrow \quad \nabla \cdot \overline{\boldsymbol{u}} = 0.$$

Result 1

$$\nabla \cdot \overline{\boldsymbol{u}} = 0 \qquad \text{(mean flow is divergence-free).}$$

STEP 3: DEDUCE DIVERGENCE-FREE FLUCTUATIONS

Recall the pointwise identity:

$$\nabla \cdot \overline{\boldsymbol{u}} + \nabla \cdot \boldsymbol{u}' = 0 \quad \text{(for all } \boldsymbol{x}, t\text{)}.$$

But we just proved $\nabla \cdot \overline{\boldsymbol{u}} = 0$ (for all $\boldsymbol{x}$), hence

$$\nabla \cdot \boldsymbol{u}' = -\nabla \cdot \overline{\boldsymbol{u}} = 0.$$

Result 2

$$\nabla \cdot \boldsymbol{u}' = 0 \qquad \text{(fluctuations are divergence-free)}.$$

INTERPRETATION: WHAT DID WE GAIN?

We started with one constraint:

$$\nabla \cdot \boldsymbol{u} = 0.$$

After Reynolds decomposition + time averaging we obtain:

$$\nabla \cdot \overline{\boldsymbol{u}} = 0, \qquad \nabla \cdot \boldsymbol{u}' = 0.$$

- This will simplify bookkeeping in the momentum derivation.
- The hard part is **not** continuity—it's what averaging does to **nonlinear products**.

PREVIEW: MOMENTUM EQUATION AND CLOSURE

Next, we will substitute

$$\boldsymbol{u} = \overline{\boldsymbol{u}} + \boldsymbol{u}', \qquad p = \overline{p} + p'$$

into

$$\frac{\partial \boldsymbol{u}}{\partial t} + (\boldsymbol{u} \cdot \nabla)\boldsymbol{u} = -\nabla p + \frac{1}{\mathrm{Re}} \Delta \boldsymbol{u}$$

and time-average.

TAKEAWAYS

- Reynolds decomposition: $\boldsymbol{u} = \overline{\boldsymbol{u}} + \boldsymbol{u}'$ with $\overline{\boldsymbol{u}'} = 0$.
- Time averaging (stationary assumption) + commutation with spatial derivatives.
- Applying to continuity yields:

$$\nabla \cdot \overline{\boldsymbol{u}} = 0, \qquad \nabla \cdot \boldsymbol{u}' = 0.$$

- Next: momentum averaging $\Rightarrow$ Reynolds stresses and the closure problem.

REFERENCES AND SCOPE

- O. Reynolds, “On the dynamical theory of incompressible viscous fluids and the determination of the criterion,” *Philosophical Transactions of the Royal Society A*, 186 (1895). Pages 123–164; doi:10.1098/rsta.1895.0004.
- S. B. Pope, *Turbulent Flows*, Cambridge University Press (2000), Chapters 3–4.
- S. L. Brunton, lecture on Reynolds-averaged Navier–Stokes equations (linked on the title slide), YouTube; the slide sequence of this deck is adapted from it.
- J. M. McDonough, *Lectures in Elementary Fluid Dynamics* (2009), source for the decomposition graphic.
- tec-science.com, “Pressure loss in pipe systems,” source for the laminar/turbulent pipe-profile graphic.

Scope of this derivation

The deck uses a long-time average for a statistically stationary, incompressible, constant-property flow. Ensemble- or phase-averaged formulations use the same Reynolds rules when the chosen operator is linear and commutes with the required derivatives.